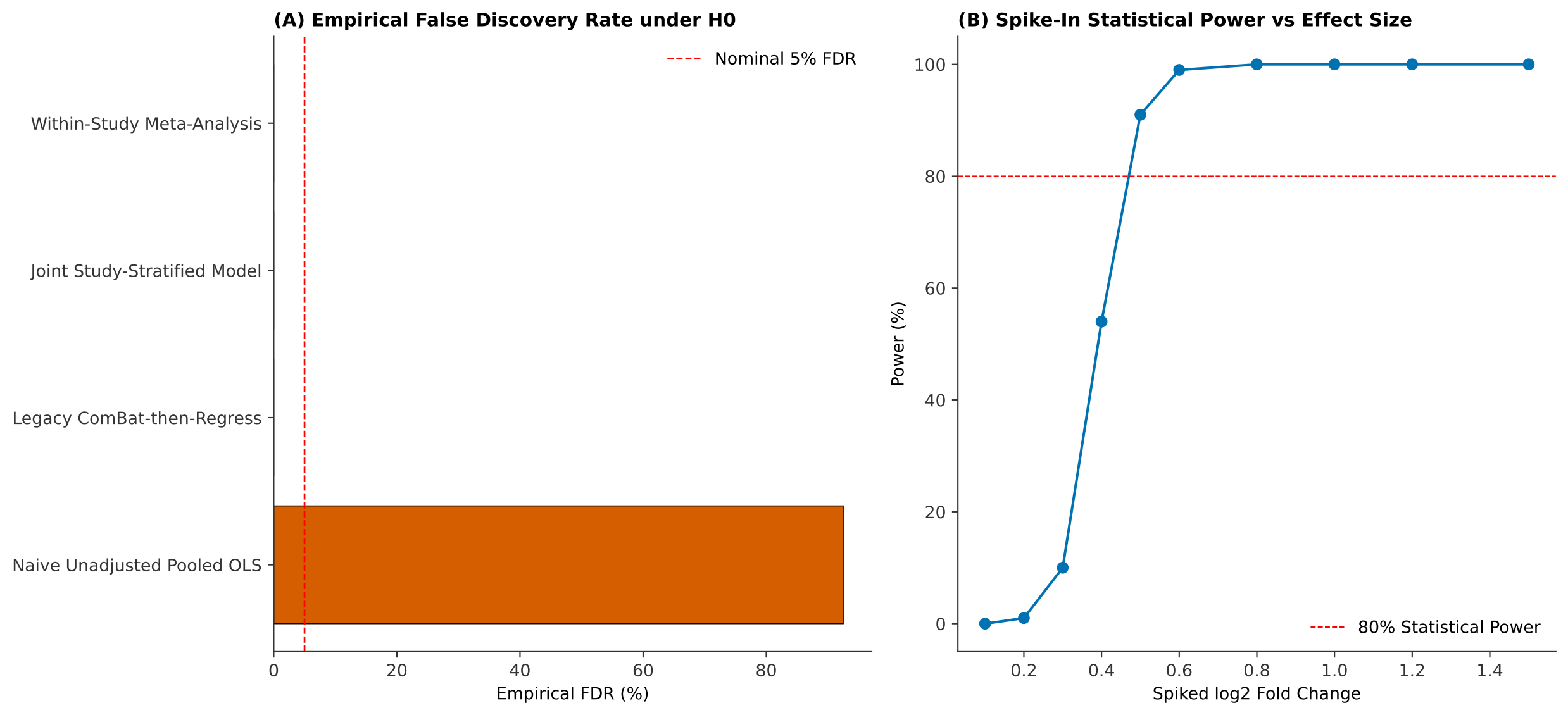
(A) Empirical False Discovery Rate under H0
Nominal 5% FDR
Within-Study Meta-Analysis
Joint Study-Stratified Model
Legacy ComBat-then-Regress
Naive Unadjusted Pooled OLS
0
20
40
60
80
Empirical FDR (%)
(B) Spike-In Statistical Power vs Effect Size
Power (%)
100
80
60
40
20
0
80% Statistical Power
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Spiked log2 Fold Change